



DATA 1301

Introduction to Data Science

Logic and Probability Theory

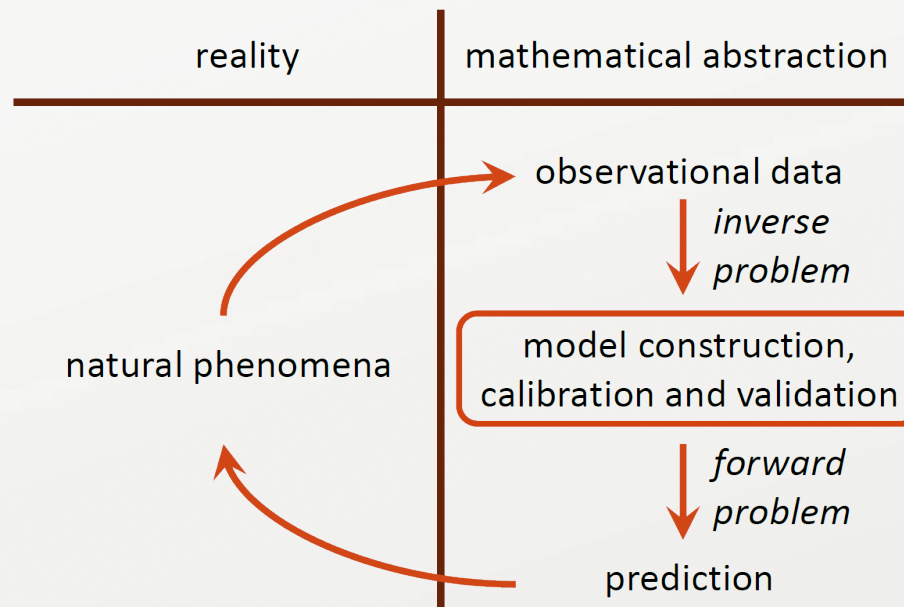
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Data Science Program / College of Science
The University of Texas
Arlington, Texas

The two classical pillars of science: Experiment and Theory

How do we make a scientific inference?

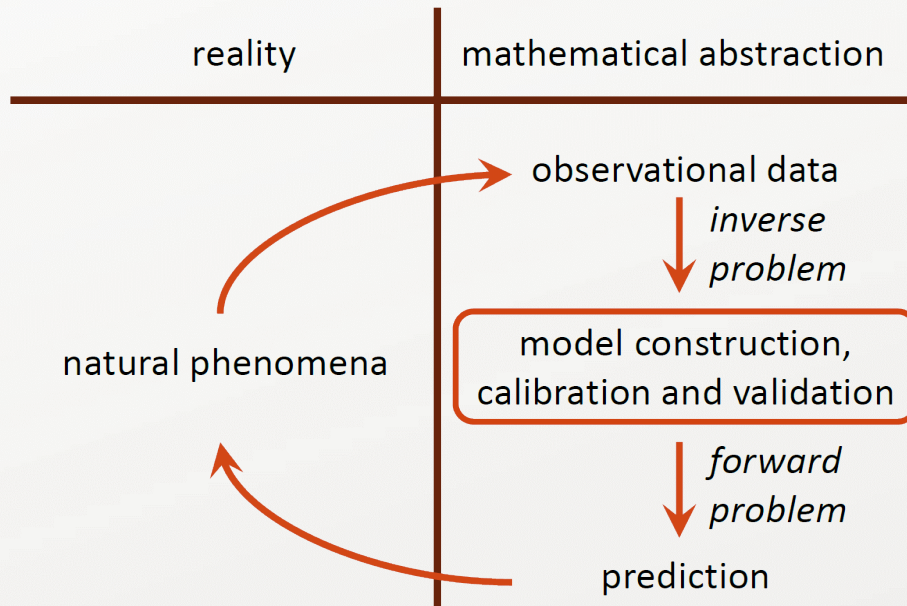
A very elementary depiction of the scientific method



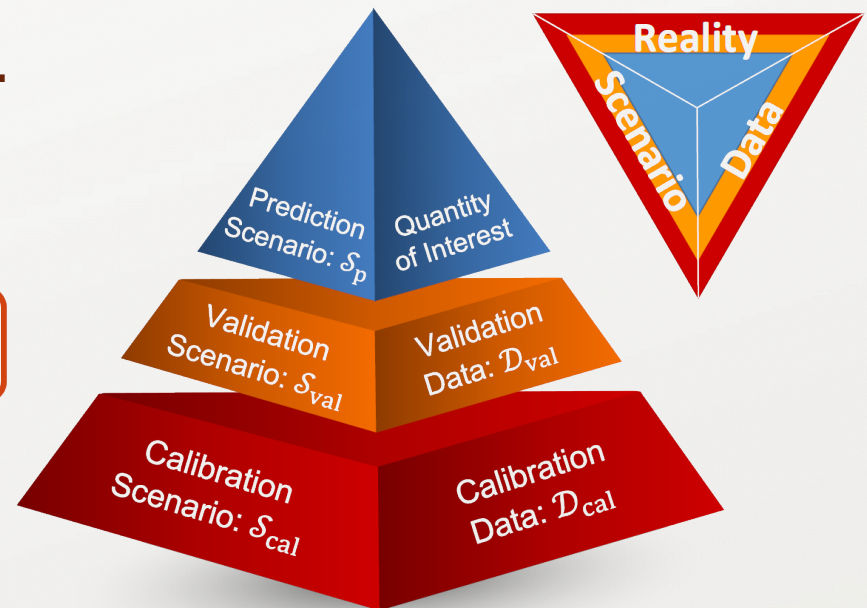
The two classical pillars of science: Experiment and Theory

How do we make a scientific inference?

A very elementary depiction of the scientific method



The prediction pyramid



The desiderata of Probability Theory

There is a set of properties that we **desire** to have in a theory of probability that we wish to construct now.

- (I) Degrees of plausibility are represented by real numbers.
- (II) Qualitative correspondence with common sense.
- (III) Consistency.
 - (I) If a conclusion can be reasoned out in more than one way, then every possible way must lead to the same result.
 - (II) We must always consider all the evidence relevant to a question. We should not arbitrarily ignore some information, basing the conclusions only on what remains.
 - (III) We must always represent equivalent states of knowledge by equivalent plausibility assignments. That is, if the state of knowledge is the same (except perhaps for labeling the propositions) in two problems, then it must assign the same plausibilities in both.

An example of correspondence with common sense

First, let's learn the conditional notation: A proposition (A) whose truth is conditioned on the truth of another proposition (B) is typically denoted by,

$$A|B$$

Second, **by convention**, we will assume that propositions with greater degree of plausibility correspond to greater real numbers.

Therefore,

$$(A|B) > (C|B)$$

An example of correspondence with common sense

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Second, **by convention**, we will assume that propositions with greater degree of plausibility correspond to greater real numbers.

Therefore,

$$(A|B) > (C|B)$$

says that, given B, A is more plausible than C.

Now, what do we mean by “correspondence with common sense” ? Suppose,

$$B|C' > B|C$$

$$A|BC' = A|BC$$

Then, the desiderata of “correspondence with commonsense” requires us to have,

$$AB|C' \geq AB|C$$

An example of correspondence with common sense

To illustrate the principle of commonsense with an example, consider the following scenario,

$A \equiv$ The probability that I will go to school today

$B \equiv$ The probability that it will rain today

$C \equiv$ The probability that the sky will be sunny today

$C' \equiv$ The probability that the sky will be cloudy today

Therefore,

$$B|C' > B|C$$

$$A|BC' = A|BC$$

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Probability is real number whose range must be either [0,1] or [1,+infinity]

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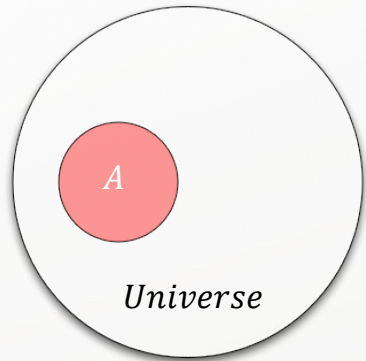
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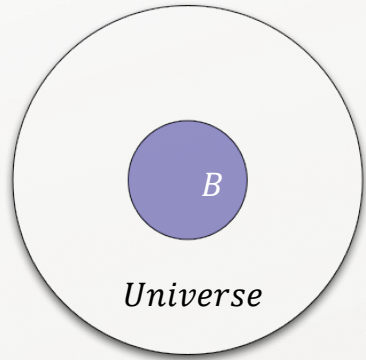
$$AB|C' \geq AB|C$$

Without going through proofs, we will state here that our three desiderates probability theory dictate that **probability is a real number whose range must be either [0,1] or [1,+infinity]**.

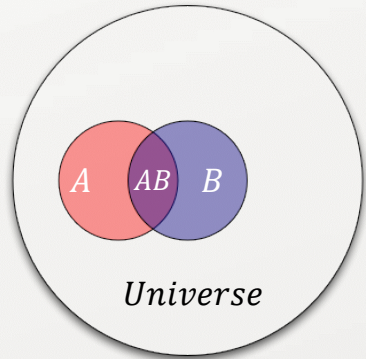
The product Rule (A prelude to the Bayesian Probability Theory)



$$P(A) = \frac{A}{U}$$



$$P(B) = \frac{B}{U}$$



$$P(AB) = \frac{AB}{U}$$

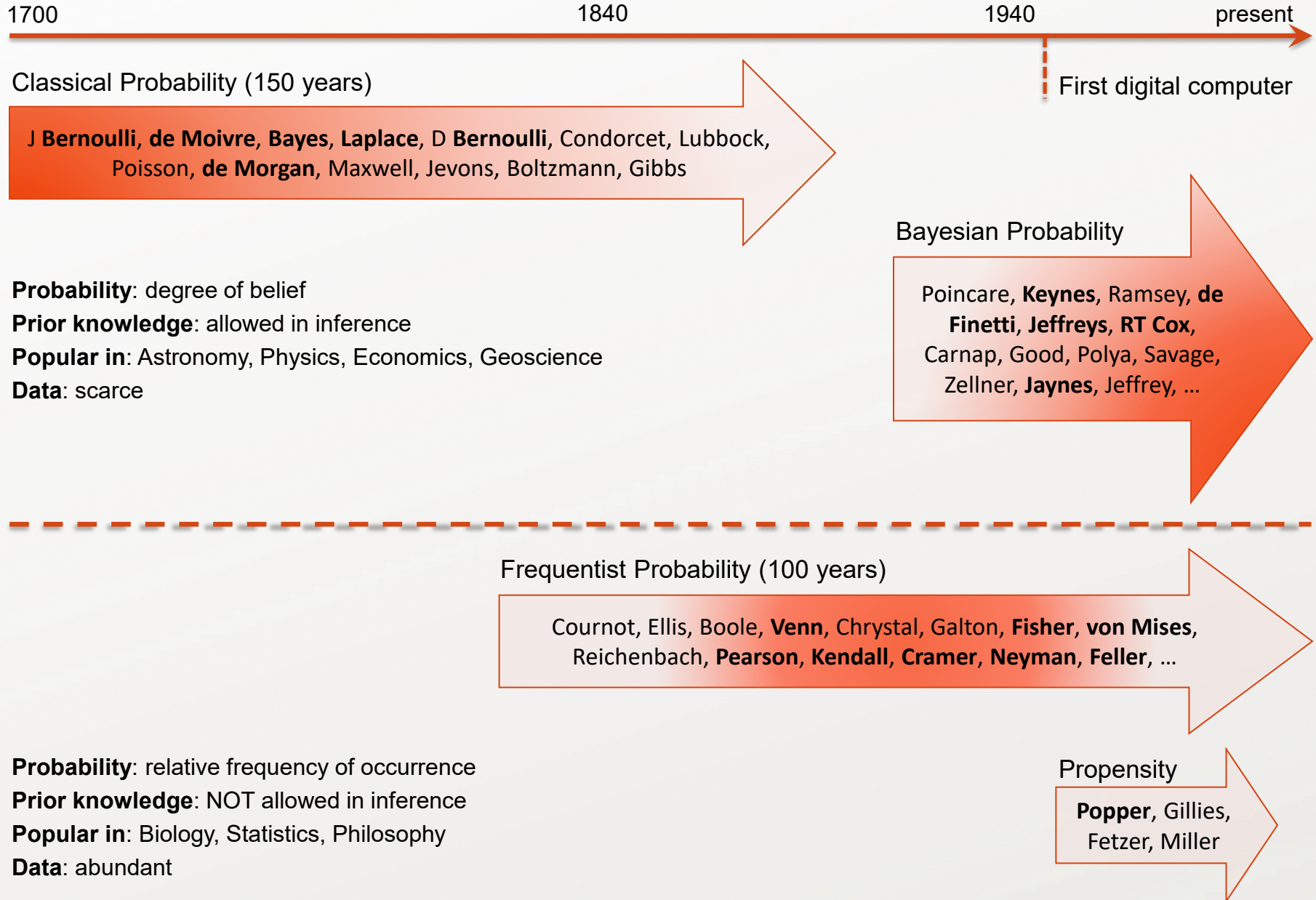
$$P(A|B) = \frac{AB}{B} = \frac{\frac{AB}{U}}{\frac{B}{U}} = \frac{P(AB)}{P(B)}$$

$$P(B|A) = \frac{AB}{A} = \frac{\frac{AB}{U}}{\frac{A}{U}} = \frac{P(AB)}{P(A)}$$

Bayes rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

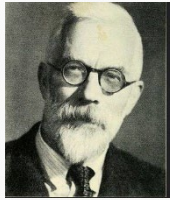
A Tug-of-War in the History of Probability Theory



Two Philosophically distinct approaches to Scientific inference

Frequentist Inference

Neyman–Pearson–Wald theory



Ronald Fisher
(1890 – 1962)

Statistician / Biologist



Jerzy Neyman
(1894 – 1981)

Statistician / Astronomer



Egon Pearson
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Abraham Wald
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No prior knowledge allowed.
Let the data speak for itself.
- R. A. Fisher

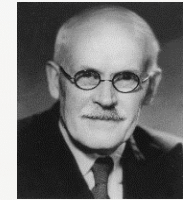
Bayesian Inference

Bayesian probability theory



Pierre Laplace
(1749 – 1827)

Astronomer / Mathematician



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Physicist



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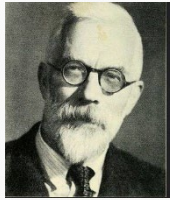
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Prior knowledge has a fundamental role
in inference along with Data
(via the Bayes' Rule)

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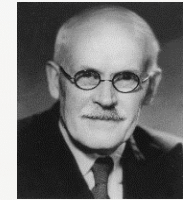
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Bayes rule:

$$\underbrace{\pi(\boldsymbol{\theta}|\mathcal{D}, M)}_{\text{posterior}} = \frac{\underbrace{\pi(\mathcal{D}|\boldsymbol{\theta}, M)}_{\text{likelihood}} \underbrace{\pi(\boldsymbol{\theta}|M)}_{\text{prior}}}{\underbrace{\pi(\mathcal{D}|M)}_{\text{evidence}}}$$

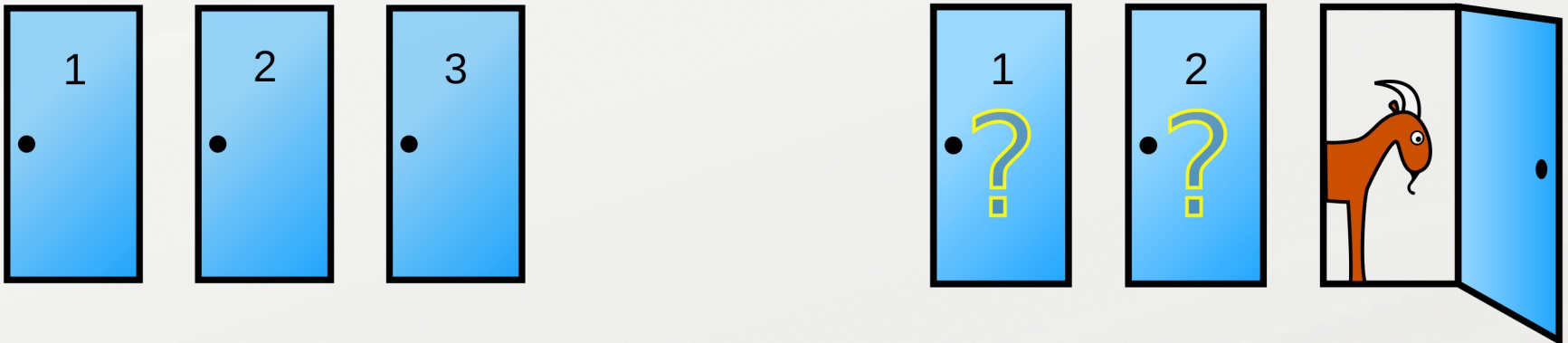
Digression: Bayes rule - The optimal method of inference

The Monty Hall Problem and Bayes Rule

Steve Selvin, 1975, "A problem in probability (letter to the editor)". *American Statistician*



Correct answer: The advantage of switching door, depends on your knowledge about the host's decision.



Digression: Bayes rule - The optimal method of inference

The Monty Hall Problem and Bayes Rule

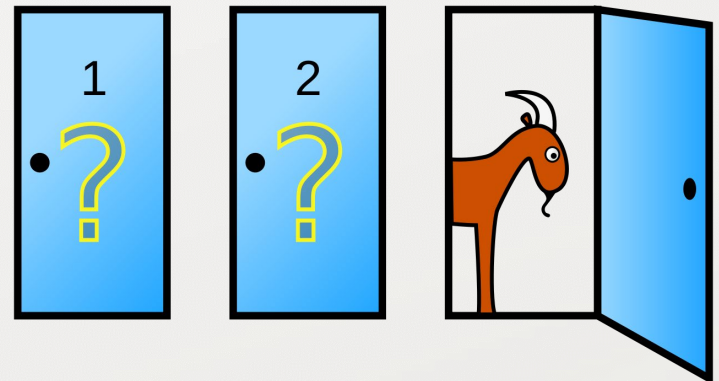
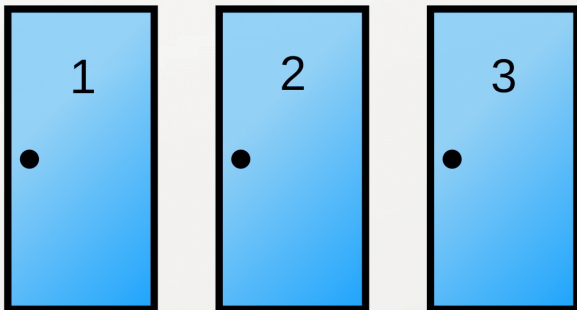
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Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the two others, goats. **You pick a door, say No. 1, and the host opens another door, say No. 3, which has a goat.** He then says to you, "Do you want to pick door No. 2?"

Question:

Is it to your advantage to switch your choice from door 1 to door 2?

Correct answer: The advantage of switching door, depends on your knowledge about the host's decision.

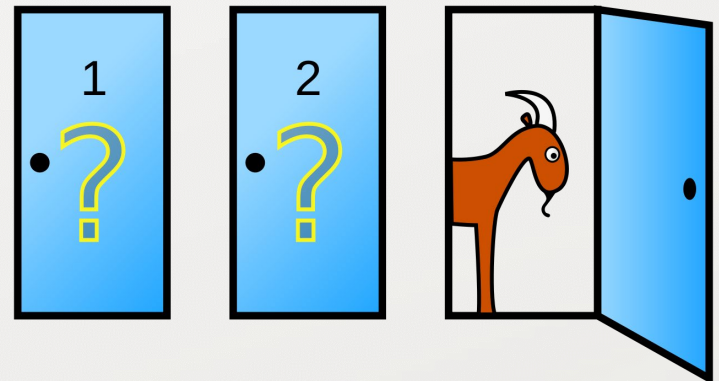
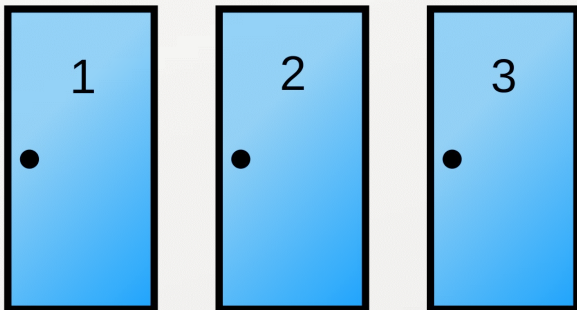


Digression: Bayes rule - The optimal method of inference

Case 1: Informed Host: Knows where the car is and will not open the door that leads to car.

You, the guest, choose door 1. The **Informed** host **consciously** opens door 3.

$$P(C2|H3, G1) = \frac{P(H3|C2, G1) P(C2)}{P(H3|G1)}$$



Digression: Bayes rule - The optimal method of inference

Case 1: Informed Host: Knows where the car is and will not open the door that leads to car.

You, the guest, choose door 1. The **Informed** host **consciously** opens door 3.

Prior knowledge about the car being behind door 2 (C2):

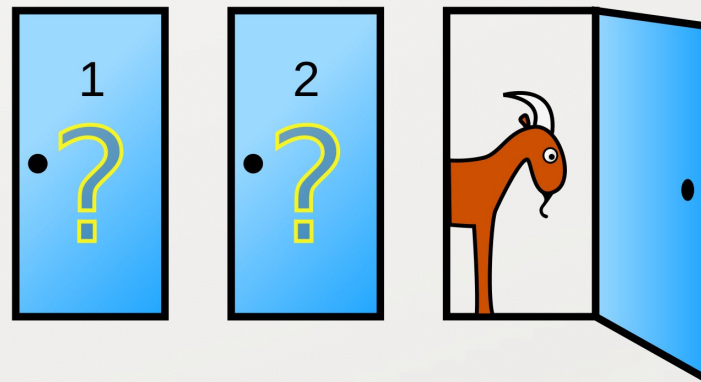
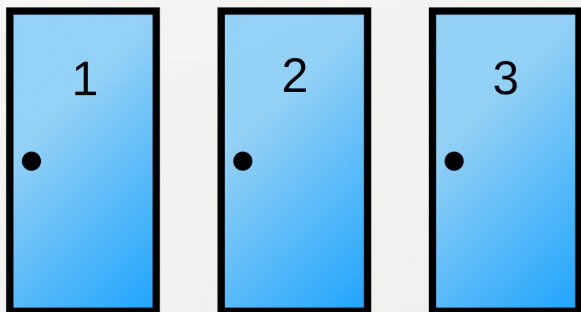
$$P(C2|G1) = P(C1|G1) = P(C3|G1) = P(C2) = \frac{1}{3}$$

New knowledge: The informed host chooses door 3 (**H3**) (He knows that the car is behind door 2)

$$P(H3|C2, G1) = 1 \qquad P(H3|G1) = \frac{1}{2}$$

Update your knowledge about door C2 using Bayes rule:

$$P(C2|H3, G1) = \frac{P(H3|C2, G1) P(C2)}{P(H3|G1)} = \frac{1 \times 1/3}{1/2} = \frac{2}{3}$$



Digression: Bayes rule - The optimal method of inference

Case 1: Informed Host: Knows where the car is and will not open the door that leads to car.

You, the guest, choose door 1. The **Informed** host **consciously** opens door 3 (knowing the car is behind #2).

Prior knowledge about the car being behind door 2 (C_2):

$$P(C_2|G_1) = P(C_1|G_1) = P(C_3|G_1) = P(C_2) = \frac{1}{3}$$

New knowledge: The informed host chooses door 3 (H_3) (He knows that the car is behind door 2)

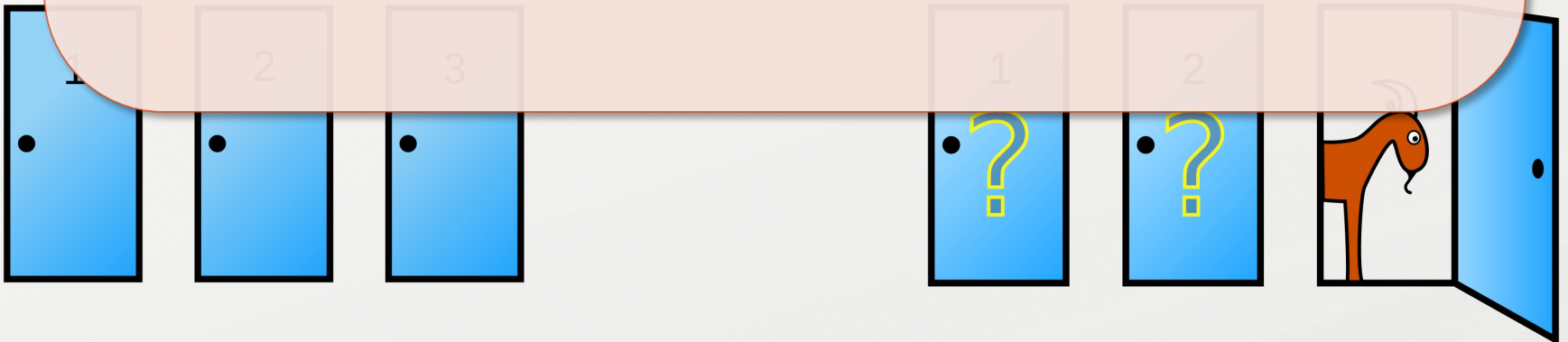
$$P(H_3|C_2, G_1) = \frac{1}{2} \quad P(C_2|H_3, G_1) = \frac{1}{2}$$

Update your knowledge about door C using Bayes rule:

$$P(C_1) = \frac{1}{3}$$

$$P(C_2 \cup C_3) = \frac{2}{3}$$

$$P(C_2|H_3, G_1) = \frac{P(H_3|C_2, G_1) \times P(C_2|G_1)}{P(H_3|G_1)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$



Digression: Bayes rule - The optimal method of inference

Case 1: Uninformed Host

You, the guest, choose door 1. The **Uninformed** host **randomly** opens door 3 (knowing nothing a priori).

Prior knowledge about the car being behind door 2:

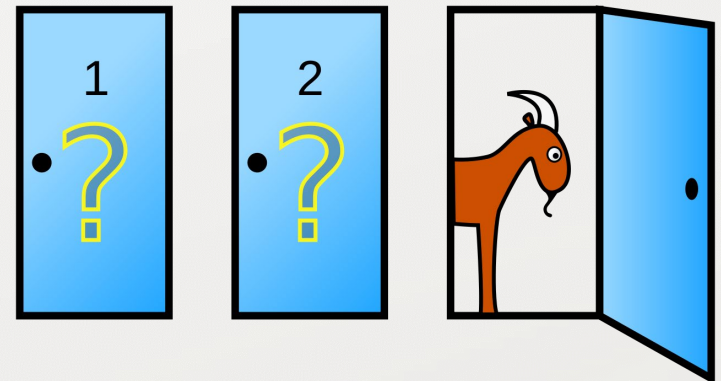
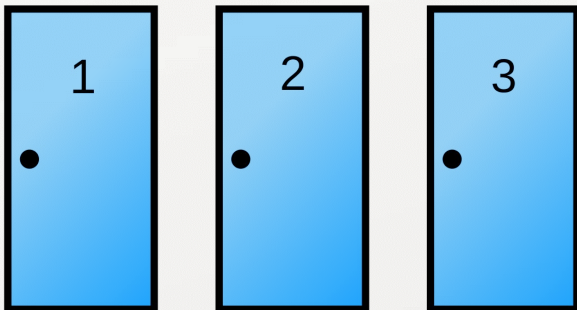
$$P(C2|G1) = P(C1|G1) = P(C3|G1) = P(C2) = \frac{1}{3}$$

New knowledge: The **Uninformed** host chooses door 3 (**H3**) (He does **not** know where the car is)

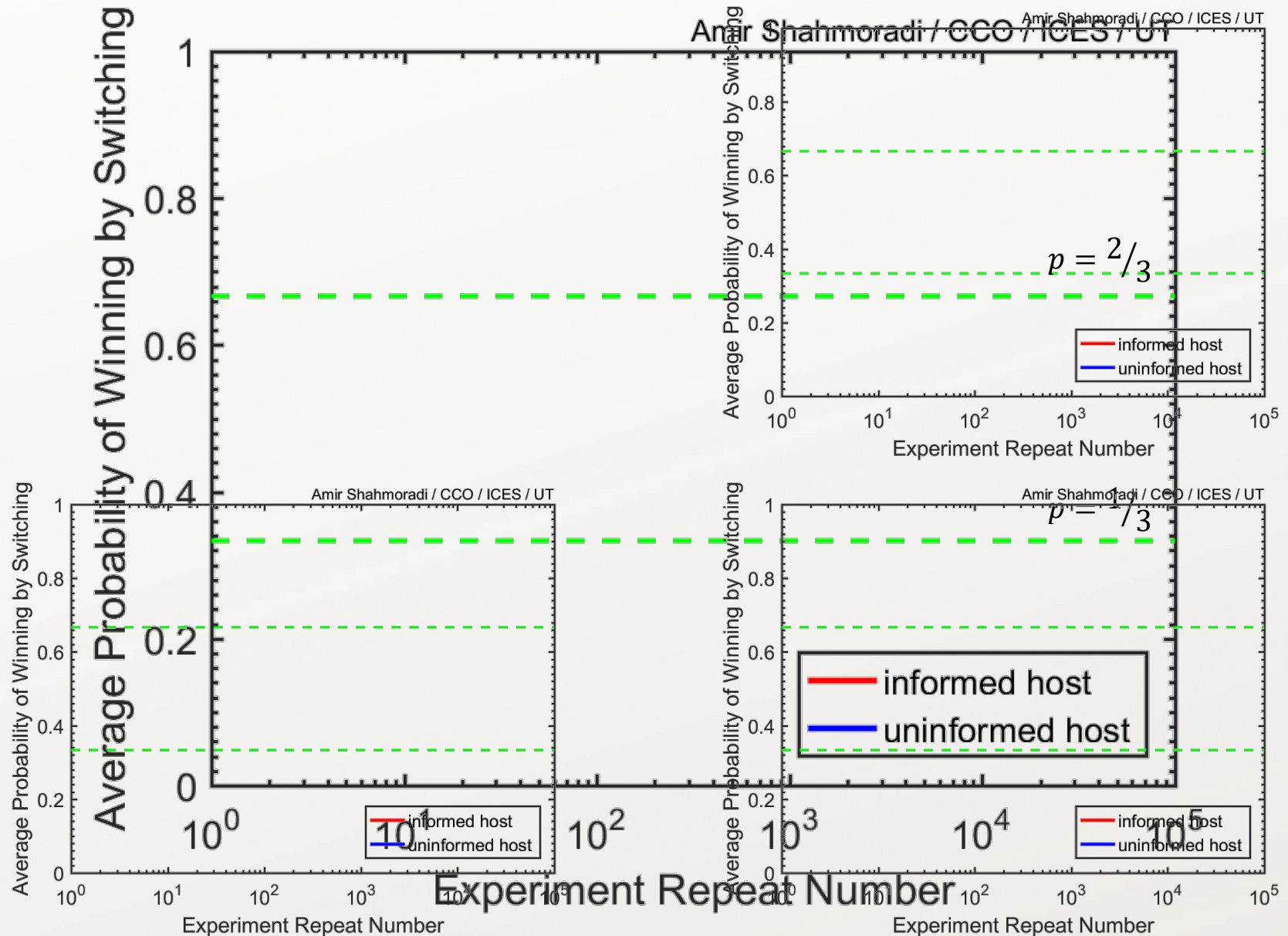
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Update your knowledge about door C2 using Bayes rule:

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Digression: Bayes rule - The optimal method of inference



There is only one type of uncertainty in the world – epistemic

Suppose you have **blurred** vision.
You throw a die **once and** read your observation (possibly wrong reading).
What is the **type of uncertainty** in your observation?

Frequentist Inference

The uncertainty is due to my **lack of knowledge**.
I **can reduce uncertainty** with better vision.
Therefore, the **uncertainty** is **epistemic**.

Bayesian Inference

The uncertainty is due to my **lack of knowledge**.
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There is only one type of uncertainty in the world – epistemic

Suppose you have **perfect** vision.

You throw a die **multiple times and** read your observations.

What is the **type of uncertainty (source of variability)** in your observations?

Frequentist Inference

The uncertainty is **inherent in the experiment**.

I **cannot reduce** uncertainty any further.

Therefore, the **uncertainty** is **aleatoric**.

Bayesian Inference

The uncertainty is due to my **lack of knowledge**:

1. Wrong / **inadequate** model.
2. Lack of sufficiently-detailed data which leads to inadequate model.

I **can reduce uncertainty** with better data / model.

Therefore, the **uncertainty** is **epistemic**.



There is only one type of uncertainty in the world – epistemic

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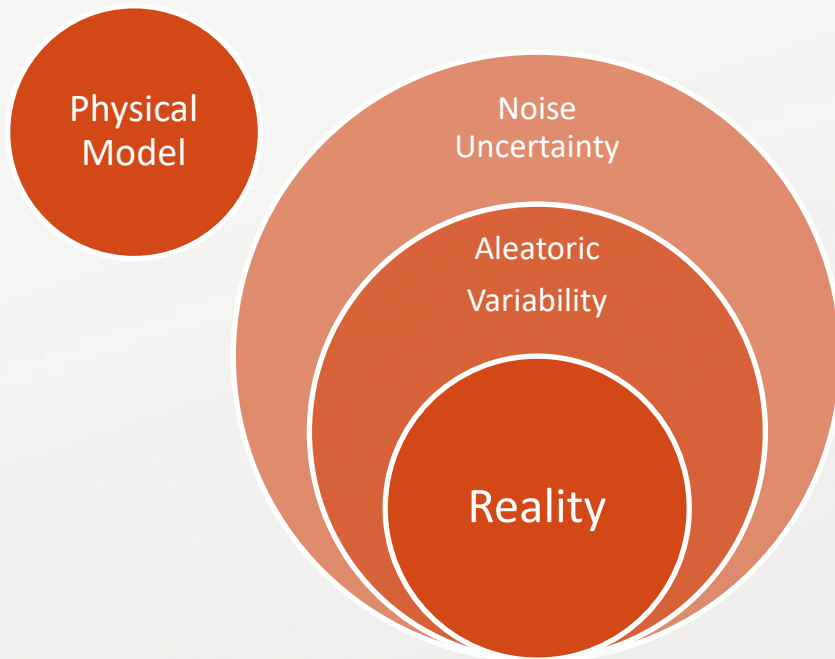
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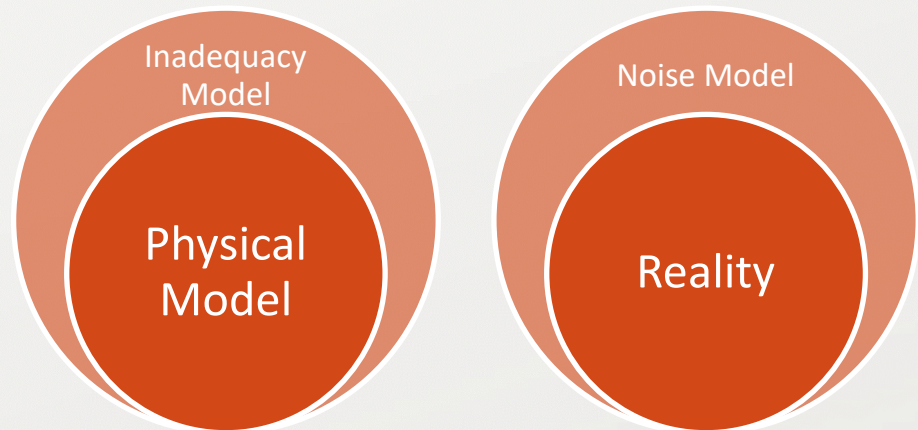
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Observations, random variables, and the likelihood principle

Suppose we observe the following data of n events, each described by m variables,

$$\mathcal{D} = \{D_1, D_2, \dots, D_n\}$$

$m = 5$ variables , or attributes, or features

$\mathcal{D}$	<i>DATA</i>	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>t</i>	ϕ
	D_1	x_1	y_1	z_1	t_1	ϕ_1
	D_2	x_2	y_2	z_2	t_2	ϕ_2
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	D_n	x_n	y_n	z_n	t_n	ϕ_n

Philosophical counterfactual assumption in statistical modeling

(John Stuart Mill, A system of Logic, Ratiocinative and Inductive, 1843)

Even though we have observed dataset $\mathcal{D}$, the experiment outcome could be different if repeated.
In other words, D is a **random variable**; a real-valued function with a **sample space** as its domain.